Appendix B. Brief comparison of concurrency libraries

Concurrency and multithreading support in programming languages and libraries aren't something new, even though standardized support in C++ is new. For example, Java has had multithreading support since it was first released, platforms that conform to the POSIX standard provide a C interface for multithreading, and Erlang provides support for message-passing concurrency. There are even C++ class libraries such as Boost that wrap the underlying programming interface for multithreading used on any given platform (whether it's the POSIX C interface or something else) to provide a portable interface across the supported platforms.

For those who are already experienced in writing multithreaded applications and would like to use that experience to write code using the new C++ multithreading facilities, this appendix provides a comparison between the facilities available in Java, POSIX C, C++ with the Boost Thread Library, and C++11, along with cross-references to the relevant chapters of this book.

Feature	Java	POSIX C	Boost threads	C++11	Chapter reference
Starting threads	java.lang.thread class	pthread_t type and associated API functions: pthread_create(), pthread_detach(), and pthread_joint ()	boost : : thread class and member functions	std : : thread class and member functions	Chapter 2
Mutual exclusion	synchronized blocks	pthread_mutex_t type and associated API functions: pthread_mutex_lockt), pthread_mutex unlock(), etc.	boost : :mutex class andmember functions, boost::lock_guard<> and boost :: unique_lock<> templates	std: :mutex class and member functions, std:: lock_guardo and std: :unique_lock<> templates	Chapter 3
Monitors/ waits for a predicate	wait() and notify () methods of the java.lang. Object class, used inside synchronized blocks	pthread_cond_t type and associated API functions: pthread_cond_wait(), pthread_cond_timed_wait ( ), etc.	boost:: condition_variable and boost:: condition_variable_any classes and member functions	std:: condition variable and std:: condition_variable any classes and member functions	Chapter 4
Atomic operations and concurrency-aware memory model	volatile variables, the types in the java.util.concurrent.atomic package	N/A	N/A	std::atomic_xxx types, std: :atomic<> class template, std::atomic_thread_fence ( ) function	Chapter 5
Thread-safe containers	The containers in the java.util.concurrentpackage	N/A	N/A	N/A	Chapters 6 and 7
Futures	java.util.concurrent.future interface and associated classes	N/A	boost :: unique_future<> and boost :: shared_future<> class templates	std: : future<>, std::shared_future<> and std::atomic_future<> class templates	Chapter 4
Thread pools	java.util.concurrent.ThreadPoolExecutor class	N/A	N/A	N/A	Chapter 9
Thread interruption	interrupt ( ) method of java.lang.Thread	pthread_cancel()	interrupt ( ) member function of boost :: thread class	N/A	Chapter 9